Improved Sidorenko bound for hypergraph Turán densities

Shuyan Chen* Xinmin Hou† Xizhi Liu‡

October 12, 2026

Abstract

For every fixed integer $r \geq 3$, we prove that the Turán density of every r-uniform hypergraph F with $m \geq 1$ edges satisfies $\pi(F) \leq 1 - c_r m^{-(r-1)/r}$, where $c_r > 0$ depends only on r. This improves Sidorenko's general bound, whose gap from 1 is of order m^{-1}. Complete r-uniform hypergraphs show that the exponent $(r-1)/r$ is optimal. The proof combines a potential argument for partial embeddings with the Lovász Local Lemma.

1 Introduction

An *r-uniform hypergraph* F, or *r-graph*, consists of a finite vertex set $V(F)$ and a set of r-element subsets of $V(F)$ called edges. All hypergraphs considered here are simple. We identify a hypergraph with its edge set, and hence write $|H|$ for the number of edges of H. Write $v(F) \coloneqq |V(F)|$.

The *Turán number* $\mathrm{ex}(n, F)$ is the maximum number of edges in an n-vertex r-graph containing no copy of F. Katona, Nemetz, and Simonovits [11] showed that the normalized Turán numbers have a limit, called the *Turán density*, $\pi(F) \coloneqq \lim_{n\to\infty} \mathrm{ex}(n, F)/\binom{n}{r}$.

We study the largest possible Turán density when the number of edges of the forbidden hypergraph is fixed. For an ordinary graph F with $m \geq 1$ edges, the theorem of Erdős and Stone [7], in the form given by Erdős and Simonovits [6], yields $\pi(F) = 1 - 1/(\chi(F) - 1)$, where $\chi(F)$ is the chromatic number of F. Since $m \geq \binom{\chi(F)}{2}$, we have $1 - \pi(F) \geq (2m)^{-1/2}$. Complete graphs show that the exponent $1/2$ is optimal.

For $r \geq 3$, a theorem of Erdős [4] shows that an r-graph F with at least one edge has $\pi(F) = 0$ if and only if it is r-partite, meaning that $V(F)$ can be partitioned into r parts so that every edge meets each part once. Positive Turán densities are much harder to determine. Even for the complete 3-graph $K_4^{(3)}$ on four vertices, the conjecture of Turán [24] that $\pi(K_4^{(3)}) = 5/9$ remains open. For further background on hypergraph Turán problems, we refer the reader to the survey of Keevash [13].

For upper bounds in terms of the number of forbidden edges, Sidorenko [21] proved that every r-graph F with $m \geq 2$ edges satisfies

$$\pi(F) \leq 1 - \frac{1}{m-1}.$$

Sidorenko's later account [22] discusses this bound and the analytic methods behind it. Keevash [12, Theorem 1(ii)] improved the bound by a lower-order term. For fixed $r \geq 3$, his result gives, uniformly over r-graphs F with m edges as $m \to \infty$,

$$\pi(F) < 1 - \frac{1}{m-1} - \frac{1 + o(1)}{2(r!)^{2/r} m^{3-2/r}}.$$

Both bounds give a gap from 1 of order m^{-1}. We improve this order to $m^{-(r-1)/r}$.

Theorem 1.1. *For every integer $r \geq 3$ and every r-graph F with $m \geq 1$ edges,*

$$\pi(F) \leq 1 - \frac{1}{4r^2} \left(\frac{4}{r}\right)^{1/r} \left(1 - \frac{1}{r}\right)^{2-2/r} \frac{1}{m^{(r-1)/r}}.$$

*Department of Mathematics, University of Manchester, Manchester M13 9PL, United Kingdom. Email: shuyan.chen-2@student.manchester.ac.uk.

†School of Mathematical Sciences, University of Science and Technology of China, Hefei 230026, China. Hefei National Laboratory, University of Science and Technology of China, Hefei 230088, Anhui, China. Email: xmhou@ustc.edu.cn.

‡School of Mathematical Sciences, University of Science and Technology of China, Hefei 230026, China. Email: liuxizhi@ustc.edu.cn.

For integers $s \geq r$, let $K_s^{(r)}$ be the complete r-graph on s vertices. De Caen [3] proved that $\pi(K_s^{(r)}) \leq 1 - \binom{s-1}{r-1}^{-1}$. Sidorenko [20] proved the lower bound $\pi(K_s^{(r)}) \geq 1 - ((r-1)/(s-1))^{r-1}$ for $s \geq r$; see also his survey [23, Section 4.2]. For fixed r, $K_s^{(r)}$ has $m = \binom{s}{r} = \Theta_r(s^r)$ edges. Since $\binom{s-1}{r-1}^{-1} = s/(rm)$, the two bounds give $1 - \pi(K_s^{(r)}) = \Theta_r(m^{-(r-1)/r})$ as $s \to \infty$.

For integers $r \geq 3$ and $m \geq 1$, define

$$\Pi_r(m) \coloneqq \max\{\pi(F) \colon F \text{ is an } r\text{-graph with } |F| = m\}.$$

It is clear that deleting isolated vertices does not change the Turán density. After their deletion, an m-edge r-graph has at most rm vertices, so only finitely many isomorphism types occur and the maximum exists.

For fixed $r \geq 3$ and any sufficiently large integer m, choose the largest integer $s \geq r$ with $\binom{s}{r} \leq m$. Since $\binom{s}{r} \leq m < \binom{s+1}{r}$, we have $m = \Theta_r(s^r)$ as $m \to \infty$. Add $m - \binom{s}{r}$ pairwise disjoint edges on new vertices to $K_s^{(r)}$, obtaining an m-edge r-graph G. Every $K_s^{(r)}$-free host is G-free, so Sidorenko's bound gives

$$1 - \pi(G) \leq 1 - \pi(K_s^{(r)}) \leq \left(\frac{r-1}{s-1}\right)^{r-1} = O_r\big(m^{-(r-1)/r}\big).$$

Combining Theorem 1.1 with the bound on $1 - \pi(G)$ above gives, for each fixed $r \geq 3$ as $m \to \infty$,

$$1 - \Pi_r(m) = \Theta_r\big(m^{-(r-1)/r}\big). \tag{1.1}$$

Keevash [12, Remark (1)] suggested that complete r-graphs should have the largest Turán density among r-graphs with the same number of edges and observed that this assertion, together with de Caen's bound, would give a gap of order $m^{-(r-1)/r}$. Theorem 1.1 establishes a lower bound of this order on $1 - \pi(F)$ for every r-graph F.

For $m = 3$, Frankl [8] proved that Sidorenko's bound $1/2$ is attained by the *expanded triangle* of every even uniformity $r = 2k \geq 4$. An expanded triangle consists of the three pairwise unions of three disjoint k-sets. Keevash and Sudakov [15, Theorem 1.1] showed that, for sufficiently large n, the extremal expanded-triangle-free r-graph consists of all r-sets meeting one part of a bipartition in an odd number of vertices, with the part sizes chosen to maximize the number of edges. Hou, Liu, Zhang, Zhao, and Zhu [10, Theorem 1.1 and Corollary 1.2] proved that every three-edge r-graph F with $r \geq 3$ has $\pi(F) < 1/2$ unless deleting its isolated vertices gives an expanded triangle, answering negatively a question of Keevash [12, Remark (2)] which asks whether Sidorenko's bound $1/2$ can be attained for odd r. Other hypergraph Turán problems with three-edge forbidden configurations include those for cancellative hypergraphs [1, 14], generalized triangles [9, 19], and weakly triangle-free hypergraphs [18, 16].

Other upper bounds depend on the intersections among the forbidden edges. Chen, Liu, Yang, and Zhu [2, Theorems 1.1 and 1.3] obtained upper bounds for the Turán densities of certain hypergraphs with prescribed edge intersections in terms of the maximum size of set families with restricted intersection sizes or a bounded matching number. Using the Lovász Local Lemma, Lu and Székely [17, Theorem 4] proved that $\pi(F) \leq 1 - 1/(\mathrm{e}(d+1))$ if every edge of F meets at most d other edges.

2 Proof of the main theorem

We prove Theorem 1.1 by combining a potential argument with the Local Lemma. We construct the embedding in two stages. In the first stage, we assign vertices whose incident unfinished edges have large total conditional failure probability, keeping all chosen vertices of the host hypergraph distinct and in a clique. One potential records the conditional failure probabilities of unfinished edges of F, and another bounds the proportion of host vertices excluded by the clique requirement. Claim 2.1 gives the estimates used in Claim 2.2 to assign one vertex. Iteration yields the partial map in Claim 2.3, with small total conditional failure probability at every remaining vertex. In the second stage, we map the remaining vertices independently and apply the Local Lemma in Claim 2.4 to ensure that every edge of F is preserved.

Proof of Theorem 1.1. Fix $r \geq 3$ and set

$$c_r \coloneqq \frac{1}{4r^2}\left(\frac{4}{r}\right)^{1/r}\left(1-\frac{1}{r}\right)^{2-2/r} = \left(\frac{(r-1)^{2r-2}}{4^{r-1}r^{4r-1}}\right)^{1/r}.$$

Let F be an r-graph with $m \geq 1$ edges, and fix $0 < q < c_r m^{-(r-1)/r}$. We will show that, for all sufficiently large n, every n-vertex r-graph H with $1 - r!|H|/n^r \leq q$ contains a copy of F.

The choice of q implies $m^{r-1}q^r < c_r^r$. Substituting the value of c_r and rearranging gives

$$\left(\frac{4r^4}{(r-1)^2}mq\right)^{r-1} < \left(\frac{4r^4}{(r-1)^2}\right)^{r-1}\frac{c_r^r}{q} = \frac{1}{r^3 q}.$$

We can therefore choose $T > 0$ satisfying

$$\frac{4r^4}{(r-1)^2}mq < T < \left(\frac{1}{r^3 q}\right)^{1/(r-1)}. \tag{2.1}$$

Put $M \coloneqq r^2 mq/(r-1)$. By the lower bound on T, we have $1/r - 4M/T > 1/r - (r-1)/r^2 = 1/r^2$. We can therefore fix β satisfying

$$\frac{1}{r^2} < \beta < \frac{1}{r} - \frac{4M}{T}. \tag{2.2}$$

Let H be an n-vertex r-graph with $q_H \coloneqq 1 - r!|H|/n^r \leq q$. Since $\beta > 1/r^2$, we can take n sufficiently large that

$$\frac{v(F)}{n} \leq \beta - \frac{1}{r^2} \quad \text{and} \quad \frac{\binom{v(F)}{2}}{n} < \left(1-\frac{1}{2r}\right)^m. \tag{2.3}$$

We construct the embedding by mapping the vertices of F to vertices of H. Once all vertices of an edge of F have been mapped, we call the edge *successful* if they are mapped to distinct vertices forming an edge of H, and say that it *fails* otherwise. To record failure for an ordered r-tuple of vertices of H, define $\gamma_r \colon V(H)^r \to \{0,1\}$ by

$$\gamma_r(x_1,\dots,x_r) \coloneqq \begin{cases} 0, & \text{if the } x_i \text{ are distinct and } \{x_1,\dots,x_r\} \in H, \\ 1, & \text{otherwise.} \end{cases}$$

Let $X_1,\dots,X_r$ be independent uniform vertices of H. Of the n^r equally likely ordered tuples, exactly $r!|H|$ are successful, so $\mathbb{E}[\gamma_r(X_1,\dots,X_r)] = 1 - r!|H|/n^r = q_H$. For $0 \leq j < r$ and $a_1,\dots,a_j \in V(H)$, define the conditional failure probability

$$\gamma_j(a_1,\dots,a_j) \coloneqq \mathbb{E}[\gamma_r(a_1,\dots,a_j,X_{j+1},\dots,X_r)].$$

Thus $\gamma_0 = q_H$. Each γ_j is symmetric in its arguments. We write $\mathbb{E}_x$ for expectation over a uniform vertex x of H. Averaging over the next coordinate gives, for $0 \leq j < r$,

$$\mathbb{E}_x[\gamma_{j+1}(a_1,\dots,a_j,x)] = \gamma_j(a_1,\dots,a_j). \tag{2.4}$$

To prepare for the Local Lemma, we will first construct a partial map from $V(F)$ to $V(H)$ such that the conditional failure probabilities of the edges incident to each unassigned vertex have small sum. We build the partial map one vertex at a time, requiring the chosen vertices of H to be distinct and to span a complete r-graph in H. Call a partial map satisfying this condition *legal*. Legality ensures that every completely assigned edge of F is successful.

Consider a legal partial map φ from $V(F)$ to $V(H)$, and let $U \subseteq V(F)$ be the set of unassigned vertices. For each $e \in F$, put $s_e \coloneqq |e \cap U|$, and call e *unfinished* if $s_e \geq 1$. For each unfinished edge e, let p_e be the probability that e fails when its unassigned vertices are mapped independently and uniformly to $V(H)$. If φ maps the assigned vertices of e to $a_1,\dots,a_{r-s_e}$, then $p_e = \gamma_{r-s_e}(a_1,\dots,a_{r-s_e})$. We measure the total remaining failure probability, counting once for each unassigned vertex of an edge, by

$$\Phi(\varphi) \coloneqq \sum_{\substack{e \in F \\ s_e \geq 1}} s_e p_e.$$

We write Φ for $\Phi(\varphi)$ when the partial map is fixed. For $v \in U$, let $\ell_v \coloneqq \sum_{e \in F\colon v \in e} p_e$ be the sum of the conditional failure probabilities of the edges of F containing v. Then $\Phi = \sum_{v \in U} \ell_v$, and the initial value is $\Phi_0 = rmq_H$.

Let $\mathbf{a} = (a_1, \dots, a_k)$ be the list of vertices of H selected by φ. Since φ is legal, these vertices are distinct and span a complete r-graph. A vertex of H outside $\{a_1, \dots, a_k\}$ is a legal choice for the next assignment precisely when it forms an edge of H with every $(r-1)$-subset of this set. We use the following sums to bound the proportion of vertices that fail this edge condition. Write $[k] \coloneqq \{1, \dots, k\}$, with $[0] = \emptyset$, and use $\binom{[k]}{j}$ for the family of its j-element subsets. Define

$$S_j(\mathbf{a}) \coloneqq \sum_{I \in \binom{[k]}{j}} \gamma_j((a_i)_{i \in I}) \quad \text{for } 0 \le j \le r-1, \qquad \text{and} \qquad \Xi(\mathbf{a}) \coloneqq \sum_{j=0}^{r-1} T^{r-1-j} S_j(\mathbf{a}).$$

Since each γ_j is symmetric, reordering its arguments does not change its value. We omit the argument $\mathbf{a}$ from $S_j(\mathbf{a})$ and $\Xi(\mathbf{a})$ when the partial map is fixed. For the empty partial map, $S_0 = q_H$ and $S_j = 0$ for $1 \le j \le r-1$, so the initial value of Ξ is $\Xi_0 = q_H T^{r-1}$. By the union bound, S_{r-1} bounds the proportion of vertices of H that fail to form an edge with some $(r-1)$-subset of $\{a_1, \dots, a_k\}$. For $1 \le j \le r-1$, appending a uniform random vertex of H to $\mathbf{a}$ gives an expected increase of S_{j-1} in S_j. Summing these expected increases with weights T^{r-1-j} gives an expected increase in Ξ of at most Ξ/T, as shown in Claim 2.1(ii).

We combine the two potentials by defining

$$\Psi \coloneqq \Phi + r^2 M \Xi = \sum_{\substack{e \in F \\ s_e \ge 1}} s_e p_e + r^2 M \sum_{j=0}^{r-1} T^{r-1-j} \sum_{I \in \binom{[k]}{j}} \gamma_j((a_i)_{i \in I}).$$

Write $\Psi_{v \mapsto x}$ for its value after a legal assignment of $v \in U$ to $x \in V(H)$, and Ψ_0 for its initial value.

We will maintain $\Psi \le M$. Since $\Phi, \Xi \ge 0$, this condition implies $\Xi \le 1/r^2$. By the union bound and (2.3), the proportion of host vertices that are not legal choices is at most $k/n + S_{r-1} \le v(F)/n + \Xi \le \beta$, so at least a proportion $1 - \beta$ of the host vertices are legal choices. To verify the initial condition, we use $q_H \le q$ and the upper bound on T in (2.1), which give

$$\Phi_0 \le rmq = \frac{r-1}{r} M, \quad \text{and} \quad r^2 M \Xi_0 \le r^2 M q T^{r-1} < \frac{M}{r}.$$

Adding these two inequalities yields $\Psi_0 < M$.

For an assignment $v \mapsto x$ with $v \in U$ and $x \in V(H)$, put $\mu_v \coloneqq \sum_{e \in F\colon v \in e} (s_e - 1) p_e$. Each edge $e \in F$ containing v has $s_e - 1$ unassigned vertices after this assignment. The newly completed edges contribute 0 to Φ, so the new contribution of the edges containing v is

$$\tau_v(x) \coloneqq \sum_{\substack{e \in F \\ v \in e,\ s_e \ge 2}} (s_e - 1) \gamma_{r - s_e + 1}\big((\varphi(u))_{u \in e \setminus U}, x\big).$$

Here the value of γ_{r-s_e+1} is the conditional failure probability of e after assigning $v \mapsto x$. The expression for $\tau_v(x)$ is defined for every $x \in V(H)$, including choices that are not legal. For a legal assignment $v \mapsto x$, the contributions of edges not containing v are unchanged. We therefore write the resulting value of Φ as

$$\Phi_{v \mapsto x} \coloneqq \sum_{\substack{e \in F \\ v \notin e,\ s_e \ge 1}} s_e p_e + \tau_v(x) = \sum_{\substack{e \in F \\ v \notin e,\ s_e \ge 1}} s_e p_e + \sum_{\substack{e \in F \\ v \in e,\ s_e \ge 2}} (s_e - 1) \gamma_{r - s_e + 1}\big((\varphi(u))_{u \in e \setminus U}, x\big).$$

Write $\Delta_x \Xi \coloneqq \Xi(\mathbf{a}, x) - \Xi(\mathbf{a})$ for the change in the clique potential. Let $A \subseteq V(H)$ be the set of legal choices for x. We first estimate the potential updates and the size of A.

Claim 2.1. *For every legal partial map and every $v \in U$, the following statements hold.*

(i) *We have $\mathbb{E}_x[\tau_v(x)] = \mu_v \le (r-1)\ell_v$.*

(ii) *We have $\Delta_x \Xi \ge 0$ for every $x \in V(H)$ and $\mathbb{E}_x[\Delta_x \Xi] \le \Xi/T$.*

(iii) *Every $x \in A$ satisfies $\Phi_{v \mapsto x} = \Phi - \ell_v - \mu_v + \tau_v(x)$.*

(iv) *The set A of legal choices satisfies $\frac{|A|}{n} \ge 1 - \frac{v(F)}{n} - \Xi$.*

Proof. For (i), since $1 \le s_e \le r$ for every $e \in F$ containing v, we have $\mu_v \le (r-1)\ell_v$. Using (2.4) in the definition of $\tau_v(x)$ and noting that edges with $s_e = 1$ have weight 0, we obtain

$$\mathbb{E}_x[\tau_v(x)] = \sum_{\substack{e \in F \\ v \in e,\ s_e \ge 2}} (s_e - 1)\mathbb{E}_x\big[\gamma_{r-s_e+1}\big((\varphi(u))_{u \in e \setminus U}, x\big)\big] = \sum_{\substack{e \in F \\ v \in e,\ s_e \ge 2}} (s_e - 1)p_e = \mu_v.$$

For (ii), appending x to $\mathbf{a}$ leaves the terms indexed by subsets of $[k]$ unchanged. The additional terms in $S_j(\mathbf{a}, x)$ are indexed by the j-subsets of $[k+1]$ containing $k+1$. Thus, for $1 \le j \le r-1$ and every $x \in V(H)$,

$$S_j(\mathbf{a}, x) - S_j(\mathbf{a}) = \sum_{I \in \binom{[k]}{j-1}} \gamma_j((a_i)_{i \in I}, x).$$

Since $S_0(\mathbf{a}, x) = S_0(\mathbf{a})$, $T > 0$, and all conditional failure probabilities are nonnegative, the definition of Ξ gives

$$\Delta_x \Xi = \sum_{j=1}^{r-1} T^{r-1-j}\big(S_j(\mathbf{a}, x) - S_j(\mathbf{a})\big) = \sum_{j=1}^{r-1} T^{r-1-j} \sum_{I \in \binom{[k]}{j-1}} \gamma_j((a_i)_{i \in I}, x) \ge 0.$$

By (2.4), averaging over x gives

$$\mathbb{E}_x[S_j(\mathbf{a}, x) - S_j(\mathbf{a})] = \sum_{I \in \binom{[k]}{j-1}} \gamma_{j-1}((a_i)_{i \in I}) = S_{j-1}(\mathbf{a}).$$

By linearity of expectation and these identities,

$$\mathbb{E}_x[\Delta_x \Xi] = \sum_{j=1}^{r-1} T^{r-1-j} S_{j-1} = \frac{\Xi - S_{r-1}}{T} \le \frac{\Xi}{T}.$$

For (iii), the contribution of the edges containing v before the assignment is $\sum_{e \in F:\ v \in e} s_e p_e = \ell_v + \mu_v$. After the assignment it is $\tau_v(x)$. All other contributions are unchanged, giving the stated formula for $\Phi_{v \mapsto x}$.

For (iv), let x be a uniform random vertex of H. If $x \notin A$, then either $x \in \{a_1, \dots, a_k\}$ or $\gamma_r((a_i)_{i \in I}, x) = 1$ for some $I \in \binom{[k]}{r-1}$. The first event has probability k/n. For each fixed $I \in \binom{[k]}{r-1}$, the event $\gamma_r((a_i)_{i \in I}, x) = 1$ has probability $\gamma_{r-1}((a_i)_{i \in I})$ by the definition of γ_{r-1}. The union bound and the definition of S_{r-1} therefore give

$$\begin{aligned} 1 - \frac{|A|}{n} = \mathbb{P}[x \notin A] &\le \mathbb{P}[x \in \{a_1, \dots, a_k\}] + \sum_{I \in \binom{[k]}{r-1}} \mathbb{P}[\gamma_r((a_i)_{i \in I}, x) = 1] \\ &= \frac{k}{n} + \sum_{I \in \binom{[k]}{r-1}} \gamma_{r-1}((a_i)_{i \in I}) = \frac{k}{n} + S_{r-1}. \end{aligned}$$

Since $k \le v(F)$ and Ξ is the sum of S_{r-1} and nonnegative terms, we obtain

$$\frac{|A|}{n} \ge 1 - \frac{k}{n} - S_{r-1} \ge 1 - \frac{v(F)}{n} - \Xi.$$

□

We now use these estimates to choose a legal assignment that decreases Ψ.

Claim 2.2. *Suppose that the current partial map is legal and $\Psi \le M$. If $v \in U$ satisfies $\ell_v \ge 1/(4r)$, then some legal choice x satisfies $\Psi_{v\mapsto x} < \Psi$.*

Proof. Since $\Phi, \Xi \ge 0$ and $\Psi \le M$, we have $\Xi \le 1/r^2$. By Claim 2.1(iv) and (2.3), using $\Xi \le 1/r^2$ and $\beta < 1/r < 1$, we obtain

$$\frac{|A|}{n} \ge 1 - \frac{v(F)}{n} - \Xi \ge 1 - \left(\beta - \frac{1}{r^2}\right) - \frac{1}{r^2} = 1 - \beta > 0.$$

By Claim 2.1(iii) and the definition of $\Delta_x\Xi$, we have, for every $x \in A$,

$$\Psi_{v\mapsto x} = \Psi - \ell_v - \mu_v + \tau_v(x) + r^2 M \Delta_x \Xi.$$

By definition, $\tau_v(x)$ is a sum of nonnegative terms for every $x \in V(H)$. Since x is uniform in $V(H)$ and $|A|/n \ge 1 - \beta > 0$, Claim 2.1(i) and (ii) give

$$\mathbb{E}_x[\tau_v(x) \mid x \in A] = \frac{1}{|A|}\sum_{x\in A} \tau_v(x) \le \frac{n}{|A|}\mathbb{E}_x[\tau_v(x)] \le \frac{\mathbb{E}_x[\tau_v(x)]}{1-\beta} = \frac{\mu_v}{1-\beta},$$
$$\mathbb{E}_x[\Delta_x\Xi \mid x \in A] = \frac{1}{|A|}\sum_{x\in A} \Delta_x\Xi \le \frac{n}{|A|}\mathbb{E}_x[\Delta_x\Xi] \le \frac{\mathbb{E}_x[\Delta_x\Xi]}{1-\beta} \le \frac{\Xi}{T(1-\beta)}.$$

Substituting these bounds into the preceding formula for $\Psi_{v\mapsto x}$ and writing the fractions with denominator $T(1-\beta)$ gives

$$\mathbb{E}_x[\Psi_{v\mapsto x} \mid x \in A] \le \Psi - \ell_v - \mu_v + \frac{T\mu_v + r^2 M\Xi}{T(1-\beta)} = \Psi - \ell_v + \frac{\beta T\mu_v + r^2 M\Xi}{T(1-\beta)}.$$

We now use $\mu_v \le (r-1)\ell_v$. The bounds $\Xi \le 1/r^2$ and $\ell_v \ge 1/(4r)$ also give $r^2 M\Xi \le M \le 4rM\ell_v$. Substituting these two estimates into the preceding bound and collecting the coefficient of ℓ_v, we obtain

$$\mathbb{E}_x[\Psi_{v\mapsto x} \mid x \in A] \le \Psi - \ell_v + \frac{\beta(r-1)T + 4rM}{T(1-\beta)}\ell_v = \Psi - \frac{(1-r\beta)T - 4rM}{T(1-\beta)}\ell_v.$$

By (2.2), we have $(1-r\beta)T - 4rM > 0$, so the coefficient of ℓ_v subtracted from Ψ is positive. Hence the conditional expectation is less than Ψ, and some $x \in A$ satisfies the claim. □

We now iterate these assignments to obtain a partial map to which the Local Lemma can be applied.

Claim 2.3. *There is a legal partial map with $\Psi < M$ such that $\ell_v < 1/(4r)$ for every unassigned vertex v.*

Proof. Starting with the empty partial map, for which $\Psi_0 < M$, apply Claim 2.2 whenever some unassigned vertex v satisfies $\ell_v \ge 1/(4r)$. Each step assigns one vertex, preserves legality, and decreases Ψ, so $\Psi < M$ throughout. The process stops after at most $v(F)$ steps. At termination, every unassigned vertex v satisfies $\ell_v < 1/(4r)$. □

Fix a partial map φ as in Claim 2.3, and let U be its set of unassigned vertices. If $U = \emptyset$, legality gives a copy of F. Otherwise, in the second stage we map the remaining vertices independently and uniformly to $V(H)$. We require only the edges of F to be preserved; the final images need not span a clique.

Claim 2.4. *Suppose that the current partial map is legal and $\ell_v < 1/(4r)$ for every $v \in U$. If the remaining vertices are mapped independently and uniformly to $V(H)$, then the probability that every edge of F is successful is at least $(1 - 1/(2r))^m$.*

Proof. For every unfinished edge e, let B_e be the event that e fails. It is determined by the nonempty set $e \cap U$ of at most r independent random variables, and $\mathbb{P}[B_e] = p_e$. We construct a dependency graph whose vertices are the events B_e. For distinct unfinished edges e, f, write $f \sim e$ if $e \cap f \cap U \ne \emptyset$, and join the corresponding events B_e and B_f. For each e, the events not joined to B_e depend only on variables outside $e \cap U$. Thus B_e is independent of the collection of its nonneighbors, as required for a dependency graph.

By the asymmetric Lovász Local Lemma of Erdős and Lovász [5], in the form stated by Lu and Székely [17, Lemma 2], the probability of avoiding all bad events is at least $\prod_e (1-x_e)$ if numbers $x_e \in [0,1)$ satisfy

$$p_e \le x_e \prod_{f \sim e} (1-x_f) \quad \text{for every unfinished edge } e.$$

Take $x_e := 2p_e$. Since $e \cap U$ is nonempty, the bound $\ell_v < 1/(4r)$ gives $0 \le x_e = 2p_e \le 2\ell_v < 1/(2r) < 1$ for every $v \in e \cap U$. Every neighbor of B_e shares an unassigned vertex with e. Summing the bounds on ℓ_v over the at most r vertices of $e \cap U$ gives

$$\sum_{f \sim e} p_f \le \sum_{v \in e \cap U} \ell_v < \frac{|e \cap U|}{4r} \le \frac{1}{4}.$$

The inequality $\prod_i (1-y_i) \ge 1 - \sum_i y_i$ for $y_i \in [0,1]$ therefore implies

$$\prod_{f \sim e} (1-x_f) \ge 1 - \sum_{f \sim e} x_f \ge \frac{1}{2}.$$

Since $x_e = 2p_e$, we have $x_e \prod_{f \sim e}(1-x_f) \ge 2p_e \cdot \frac{1}{2} = p_e$, which verifies the Local Lemma hypothesis. There are at most m unfinished edges, so the Local Lemma yields

$$\mathbb{P}\left[\bigcap_e \overline{B_e}\right] \ge \prod_e (1-x_e) \ge \left(1 - \frac{1}{2r}\right)^m.$$

Every completely assigned edge was already successful by legality, which proves the claim. □

To obtain a copy of F from Claim 2.4, we must also exclude repeated images at vertices that do not lie in a common edge. Since φ is legal, it is injective. For any pair of distinct vertices of F with at least one vertex in U, their images agree with probability $1/n$. By the union bound, the probability of any repeated image is at most $\binom{v(F)}{2}/n$. An assignment embeds F precisely when it is injective and every edge is successful. By Claim 2.4 and the second inequality in (2.3), we have

$$\begin{aligned}&\mathbb{P}[\text{the final map embeds } F] \\ &\quad \ge \mathbb{P}[\text{every edge is successful}] - \mathbb{P}[\text{the final map is not injective}] \ge \left(1 - \frac{1}{2r}\right)^m - \frac{\binom{v(F)}{2}}{n} > 0.\end{aligned}$$

Thus H contains F, proving the embedding assertion.

For every fixed $0 < q < c_r m^{-(r-1)/r}$, the embedding assertion gives $1 - r!\,\mathrm{ex}(n,F)/n^r > q$ for all sufficiently large n. Since $r!\,\mathrm{ex}(n,F)/n^r \to \pi(F)$, letting $n \to \infty$ and then taking the supremum over $0 < q < c_r m^{-(r-1)/r}$ gives $\pi(F) \le 1 - c_r m^{-(r-1)/r}$. □

Acknowledgements

X.H. was supported by the National Key R&D Program of China (2023YFA1010203), the National Natural Science Foundation of China (12471336). the Quantum Science and Technology-National Science and Technology Major Project (2021ZD0302902). X.L. was supported by the Excellent Young Talents Program (Overseas) of the National Natural Science Foundation of China. This research was carried out while the first author was visiting the School of Mathematical Sciences at the University of Science and Technology of China.

Declaration on the use of AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition.

References

[1] B. Bollobás. Three-graphs without two triples whose symmetric difference is contained in a third. *Discrete Math.*, 8(1):21–24, 1974.

[2] Y. Chen, X. Liu, N. Yang, and T. Zhu. Upper bounds on Turán densities via extremal set theory, 2026. Preprint, arXiv:2606.02135.

[3] D. de Caen. Extension of a theorem of Moon and Moser on complete subgraphs. *Ars Combin.*, 16:5–10, 1983.

[4] P. Erdős. On extremal problems of graphs and generalized graphs. *Israel J. Math.*, 2:183–190, 1964.

[5] P. Erdős and L. Lovász. Problems and results on 3-chromatic hypergraphs and some related questions. In *Infinite and Finite Sets, Vol. II (Colloq., Keszthely, 1973)*, volume 10 of *Colloq. Math. Soc. János Bolyai*, pages 609–627. North-Holland, Amsterdam, 1975.

[6] P. Erdős and M. Simonovits. A limit theorem in graph theory. *Studia Sci. Math. Hungar.*, 1:51–57, 1966.

[7] P. Erdős and A. H. Stone. On the structure of linear graphs. *Bull. Amer. Math. Soc.*, 52:1087–1091, 1946.

[8] P. Frankl. Asymptotic solution of a Turán-type problem. *Graphs Combin.*, 6(3):223–227, 1990.

[9] P. Frankl and Z. Füredi. Extremal problems whose solutions are the blow-ups of the small Witt-designs. *J. Combin. Theory Ser. A*, 52(1):129–147, 1989.

[10] J. Hou, X. Liu, Y. Zhang, H. Zhao, and T. Zhu. Odd hypergraph Mantel theorems, 2025. Preprint, arXiv:2510.10590.

[11] G. O. H. Katona, T. Nemetz, and M. Simonovits. On a graph-problem of Turán. *Mat. Lapok*, 15:228–238, 1964. In Hungarian.

[12] P. Keevash. The Turán problem for hypergraphs of fixed size. *Electron. J. Combin.*, 12(1):Note 11, 6 pp., 2005.

[13] P. Keevash. Hypergraph Turán problems. In *Surveys in Combinatorics 2011*, volume 392 of *London Math. Soc. Lecture Note Ser.*, pages 83–140. Cambridge Univ. Press, Cambridge, 2011.

[14] P. Keevash and D. Mubayi. Stability theorems for cancellative hypergraphs. *J. Combin. Theory Ser. B*, 92(1):163–175, 2004.

[15] P. Keevash and B. Sudakov. On a hypergraph Turán problem of Frankl. *Combinatorica*, 25(6):673–706, 2005.

[16] X. Liu. On a hypergraph Mantel theorem, 2025. Preprint, arXiv:2501.19229.

[17] L. Lu and L. Székely. Using Lovász local lemma in the space of random injections. *Electron. J. Combin.*, 14(1):Research Paper 63, 13 pp., 2007.

[18] D. Mubayi, O. Pikhurko, and B. Sudakov. Hypergraph Turán problem: some open questions. AIM workshop problem collection, 2011.

[19] O. Pikhurko. An exact Turán result for the generalized triangle. *Combinatorica*, 28(2):187–208, 2008.

[20] A. Sidorenko. Systems of sets that have the T-property. *Moscow Univ. Math. Bull.*, 36(5):22–26, 1981. English translation.

[21] A. Sidorenko. Extremal combinatorial problems in spaces with continuous measure. *Issled. Operatsiĭ i ASU*, 34:34–40, 1989. In Russian.

[22] A. Sidorenko. An analytic approach to extremal problems for graphs and hypergraphs. In *Extremal Problems for Finite Sets (Visegrád, 1991)*, volume 3 of *Bolyai Soc. Math. Stud.*, pages 423–455. János Bolyai Math. Soc., Budapest, 1994.

[23] A. Sidorenko. What we know and what we do not know about Turán numbers. *Graphs Combin.*, 11(2):179–199, 1995.

[24] P. Turán. Eine Extremalaufgabe aus der Graphentheorie. *Mat. Fiz. Lapok*, 48:436–452, 1941. In Hungarian, with German summary.